

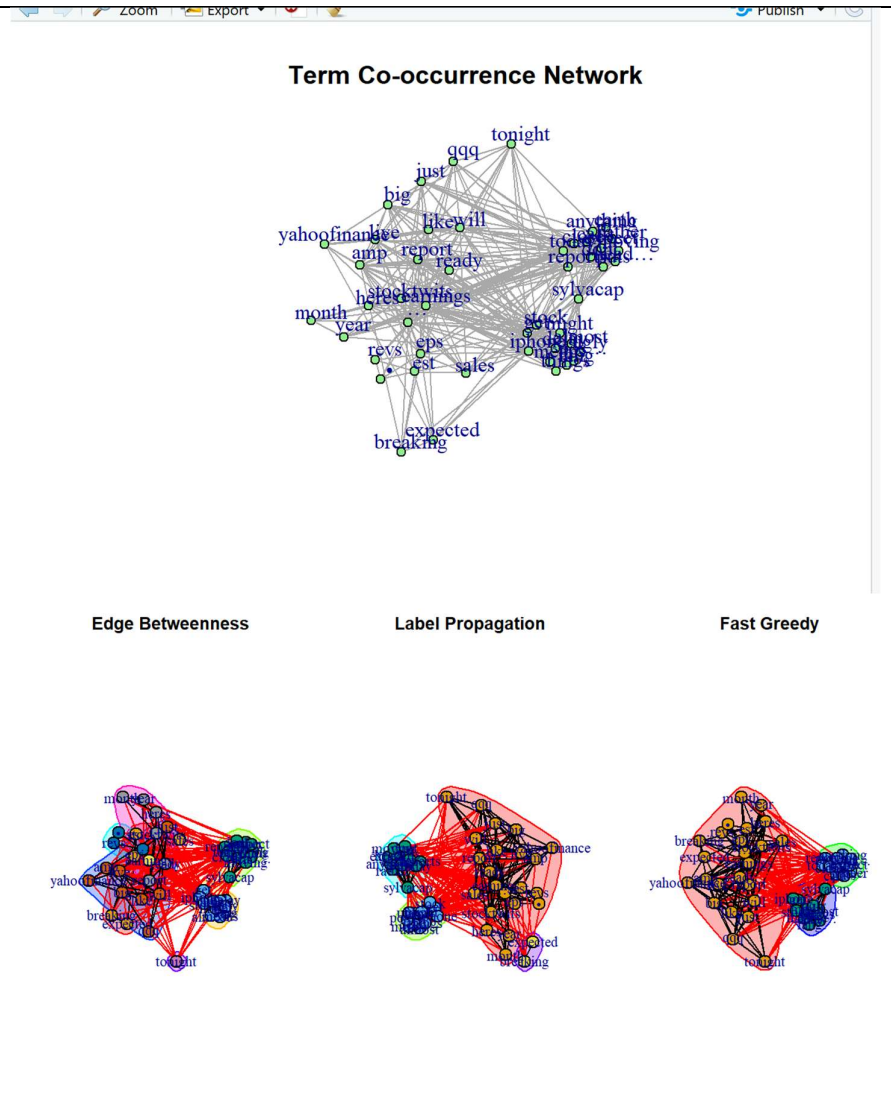
Semester	S.E. Semester IV – CMPN
Subject	R-Programming
Subject Professor In-charge	Prof. Sanjeev D.
Assisting Teachers	Prof. Prakash P.
Laboratory	M411A

Student Name	Chetan katkar
Roll Number	23102A0024
Grade and Subject Teacher's Signature	

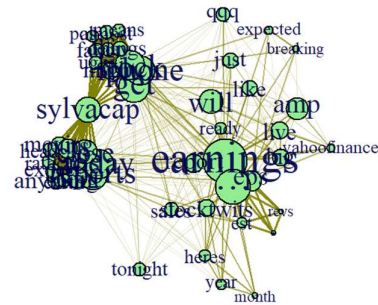
Experiment Number	05
Experiment Title	Social Network Analysis (SNA) and Co-occurrence Network Visualization using R
Objectives (Skill Set / Knowledge Tested / Imparted)	To understand and implement Social Network Analysis (SNA) workflows in R using appropriate network analysis libraries (such as igraph and tm); to import unstructured text data, preprocess and convert it into term-document co-occurrence structures, represent complex relationships as graph structures consisting of nodes (vertices) and edges (links), evaluate node centrality measures (Hubs, Authorities, and Node Degrees), perform community detection algorithm comparisons (Edge Betweenness, Label Propagation, Fast Greedy), generate dynamic multi-layout network visualizations, and maintain project version control using Git and GitHub.
Description	In this practical, an end-to-end Social Network Analysis (SNA) project was implemented in RStudio using the tm (text mining) and igraph network analysis packages. The dataset (apple.csv) containing textual records was imported, cleaned, and standardized by converting text to lower case, removing punctuation, numbers, stopwords, URLs, and specific target terms ("apple", "aapl"). A Term-Document Matrix (TDM) was generated to represent word occurrences across documents.

	The matrix was converted into adjacency structures to construct two distinct networks: 1. Term Co-occurrence Network: Representing relationship graphs between key vocabulary terms based on co-occurrence density. 2. Tweet/Document Similarity Network: Representing graph connections between documents based on shared term profiles. Structural network characteristics were evaluated by computing node degree distributions, visual degree histograms, and relative vertex connections. Network centrality and entity prominence were calculated using Hub and Authority scoring algorithms (hub_score and authority_score) to identify influential nodes. Additionally, community detection algorithms—including Edge Betweenness, Label Propagation, and Fast Greedy—were executed to identify sub-group clusters and modular structures within the graph. Network diagrams were rendered using dynamic vertex sizing based on degree metrics, customized edge weighting, and pruned layouts to isolate dense components. The complete script, datasets, and generated network visual assets were tracked using Git version control and pushed to GitHub.
--	--

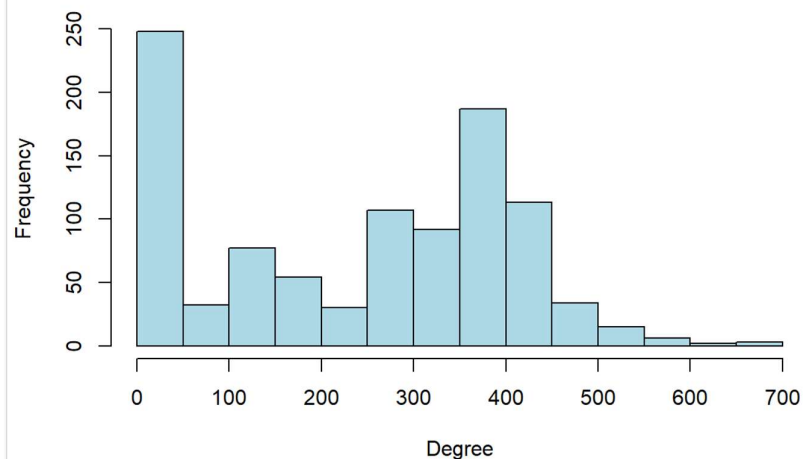
Output



Weighted Term Co-occurrence Network



Histogram of Tweet Node Degrees



Conclusion

Thus, Social Network Analysis (SNA), term co-occurrence graph construction, community detection clustering, centrality score evaluation (Hubs and Authorities), and network visualization were successfully implemented using R Programming. The practical provided hands-on experience in converting unstructured textual data into graph-theoretic adjacency structures, analyzing topological network characteristics, identifying influential network nodes, and demonstrating an integrated end-to-end R-based data analytics workflow

